



मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान भोपाल, भोपाल – 462003

Maulana Azad National Institute of Technology Bhopal, Bhopal – 462003

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Department of Computer Science & Engineering

B.Tech. VI Semester (EE1)

Data Structures (CSE352)

Assignment-IV

1. Consider the following binary tree:

```
      A
     /\
    B C
   /\ /\
  D E F G
   /\
  H I
```

- Provide the Preorder, Inorder, and Postorder traversals for this tree.
  - What is the height of the above tree?
  - Delete node E and redraw the tree. What changes occur in the **Preorder** and **Inorder** traversal outputs?
2. Given the Preorder and Inorder traversal of a Binary Tree, construct the Binary Tree  
Inorder: [40, 20, 50, 10, 60, 30], Preorder: [10, 20, 40, 50, 30, 60]
3. Find the **Preorder traversal** when the Postorder and Inorder traversal of a Binary Tree are given as [4, 2, 5, 1, 3, 6] and 4 5 2 6 3 1
4. What is the time complexity of (i) insertion (b) deletion operations in a BST and AVL tree?
5. Consider the following AVL tree:

```
      50
     /\
    30 70
   /\ /\
  20 40 60 80
```

- Add node 90
- Add node 95
- Add node 25
- Add node 22